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VISHWAKARMA INSTITUTE OF TECHNOLOGY
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Department of Engineering, Sciences and Humanities
Calculus and Statistics HS1076
UNIT V – DESCRIPTIVE STATISTICS

Tutorial 9

- 1) Construct the ungrouped frequency, relative frequency and cumulative frequency distribution table for the following data. Further find the mode, median and mean of these data sets.
 - a) 4, 3, 6, 5, 2, 4, 3, 3, 6, 4, 2, 3, 2, 2, 3, 3, 4, 5, 6, 4, 2, 3, 4
 - b) 6, 7, 5, 4, 5, 6, 6, 8, 7, 9, 6, 5, 6, 7, 7, 8, 9, 4, 6, 7, 6, 5
- 2) The marks obtained out of 25 by 30 students of a class in the examination are given below.
20, 6, 23, 19, 9, 14, 15, 3, 1, 12, 10, 20, 13, 3, 17, 10, 11, 6, 21, 9, 6, 10, 9, 4, 5, 1, 5, 11, 7, 24
Represent the above data as a grouped data taking the class interval 0 – 5
- 3) Weekly pocket expenses (in \$) of 30 students of class VIII are 37, 41, 39, 34, 71, 26, 56, 61, 58, 79, 83, 72, 64, 39, 75, 39, 37, 59, 57, 37, 53, 38, 49, 45, 70, 82, 44, 37, 79, 76.
Construct the grouped frequency table with the class interval of equal width such as 30 - 35. Also, find the range of the weekly pocket expenses.
- 4) Pulse rate (per minute) of 25 people were recorded as
61, 75, 71, 72, 70, 65, 77, 72, 67, 80, 77, 62, 71, 74, 79, 67, 80, 77, 62, 71, 74, 61, 70, 80, 72, 59, 78, 71, 72.
Construct a frequency table expressing the data in the inclusive form taking the class interval 61-65 of equal width
- 5) The frequency distribution of weights (in kg) of 40 persons is given below.

Weights (in kg)	30 - 35	35 - 40	40 - 45	45 - 50	50 - 55
Frequency	6	13	14	4	3

- (a) What is the lower limit of fourth-class interval?
 - (b) What is the class size of each class interval?
 - (c) Which class interval has the highest frequency?
 - (d) Find the class marks of all the class intervals?
- 6) Construct the frequency distribution table for the data on heights (cm) of 20 boys using the class intervals 130 - 135, 135 - 140 and so on.
The heights of the boys in cm are: 140, 138, 133, 148, 160, 153, 131, 146, 134, 136, 149, 141, 155, 149, 165, 142, 144, 147, 138, 139. Also, find the range of heights of the boys.
- 7) Construct a frequency distribution table for the following weights (in gm) of 30 oranges using the equal class intervals, one of them is 40-45 (45 not included).
The weights are: 31, 41, 46, 33, 44, 51, 56, 63, 71, 71, 62, 63, 54, 53, 51, 43, 36, 38, 54, 56, 66, 71, 74, 75, 46, 47, 59, 60, 61, 63.
 - (a) What is the class mark of the class intervals 50-55?

- (b) What is the range of the above weights?
 (c) How many class intervals are there?
 (d) Which class interval has the lowest frequency?

8) The following observations are arranged in ascending order. The median of the data is 25 find the value of x .
 17, x , 24, $x + 7$, 35, 36, 46

9) The mean of the following distribution is 26. Find the value of p and the value of the observation.

x_i	0	1	2	3	4	5
f_i	3	3	p	7	$p - 1$	4

Also, find the mode and the given data.

10) The mean of 8, 11, 6, 14, x and 13 is 66. Find the value of the observation x .

11) The mean of 6, 8, $x + 2$, 10, $2x - 1$, and 2 is 9. Find the value of x and also the value of the observation in the data.

12) The time taken in minutes to assemble a device is measured 50 times and the results are as shown.
 Draw a histogram depicting this data and determine the mean, median and modal values of the distribution.

Time	14.5-15.5	16.5-17.5	18.5-19.5	20.5-21.5	22.5-23.5	24.5-25.5
Frequency	5	8	16	12	6	3

13) The frequency distribution given below refers to the heights in centimeters of 100 people.
 Plot the histogram and determine the mean value of the distribution, correct to the nearest millimeter.

Height	150-156	157-163	164-170	171-177	178-184	185-191
Frequency	5	18	20	27	22	8

14) The gain of 90 similar transistors is measured and the results are as shown.

Energy gain	83.5-85.5	86.5-88.5	89.5-91.5	92.5-94.5	95.5-97.5
Frequency	6	39	27	15	3

Draw the histogram of this frequency distribution.

Also find the mean, median and modal values of the distribution.

15) The diameters, in centimeters, of 60 holes bored in engine castings are measured and the results are as shown.
 Draw a histogram depicting these results and further determine the mean, median and modal values of the distribution.

2.011-2.014	2.016-2.019	2.021-2.024	2.026-2.029	2.031-2.034
7	16	23	9	5

